

Reactive SDN for Securing ICS Environments

Curated Reading List, Literature-Gap Analysis & Innovation Directions

Deep literature sweep across 10 sub-domains · 40+ sources · July 2026

How to use this document

This is a prioritized, annotated map of the research most relevant to building a reactive SDN system for ICS security. Each entry is tagged **[Essential]** (read first — foundational or closest to our objective), **[High]** (strong, directly applicable), or **[Reference]** (breadth, datasets, or context). Papers are grouped into ten themes. After the reading list, a gap analysis contrasts what the field already does well against the white space our project can own, followed by concrete innovation directions and a suggested reading order.

Bottom line up front: the market and literature are saturated with ever-more-accurate *detection*, but thin on *safe, automated, device-aware network response* that provably does not endanger the physical process. That response gap — plus real-time closed-loop evaluation and controller resilience — is where an innovative, defensible contribution lies.

Theme 1 — Foundations & surveys (start here)

These frame the whole problem space and let you position the project quickly.

1. Software-Defined Networking approaches for intrusion response in Industrial Control Systems: A survey **[Essential]**

X. Etxezarreta, I. Garitano, M. Iturbe, U. Zurutuza — *Int. J. Critical Infrastructure Protection*, 42:100615, 2023.

Why it matters: The anchor reference. Provides the four-family taxonomy (dynamic filtering, survivability, MTD, honeypots), maps ICS security requirements to SDN, and lists open challenges — the exact scaffolding for our project.

Link: <https://www.sciencedirect.com/science/article/pii/S1874548223000288>

2. Detection and mitigation of cyber-attacks in SDN using ML/DL: a systematic literature review, challenges and future directions **[Essential]**

Springer, *International Journal of Information Security*, 2025.

Why it matters: Current, systematic map of ML/DL for SDN security with an explicit future-directions section. Best single source for understanding where the AI-for-SDN field is and what remains unsolved.

Link: <https://link.springer.com/article/10.1007/s10207-025-01114-z>

3. Machine learning in ICS security: current landscape, opportunities and challenges **[High]**

Springer, *Journal of Intelligent Information Systems*, 2022.

Why it matters: Grounds the ICS-specific constraints (data scarcity, safety, real-time) that make naive ML transplants fail. Good for motivating why response — not just detection — is the hard part.

Link: <https://link.springer.com/article/10.1007/s10844-022-00753-1>

4. Guide to Industrial Control Systems (ICS) Security (NIST SP 800-82 Rev. 2) **[Reference]**

K. Stouffer et al., *NIST Special Publication 800-82 Rev. 2*, 2015 (Rev. 3 draft newer).

Why it matters: The canonical requirements document. Cite it for the availability/integrity-over-confidentiality priority and the mandate that controls must not disrupt the process.

Link: <https://nvlpubs.nist.gov/nistpubs/specialpublications/nist.sp.800-82r2.pdf>

Theme 2 — SDN dynamic response & incident handling for ICS

The core mechanism of the project: alert-to-action pipelines that preserve the process.

5. Leveraging Software-Defined Networking for Incident Response in Industrial Control Systems [Essential]

A. F. M. Piedrahita, V. Gaur, J. Giraldo, Á. A. Cárdenas, S. J. Rueda — *IEEE Software*, 35(1):44–50, 2018.

Why it matters: The closest prior work to our objective: IDS + SDN/NFV substitutes estimated values for anomalous sensor readings so the process keeps running safely. The template for safety-preserving response.

Link: <https://www.scitepress.org/Papers/2019/73595/73595.pdf>

6. Enabling Dynamic Network Access Control with Anomaly-based IDS and SDN [High]

ACM Int. Workshop on Security in SDN & NFV (SDN-NFV Sec), 2019.

Why it matters: Clean formalization of the IDS→controller→flow-rule loop and the allow/block/mirror/redirect action set our system must implement.

Link: <https://dl.acm.org/doi/10.1145/3309194.3309199>

7. Securing ICS networks: SDN-based Automated Traffic Control and MTD Defensive Framework against DDoS attacks [High]

Computer Communications, Elsevier, 2025.

Why it matters: Recent, combines automated SDN traffic control with MTD for ICS — a working example of orchestrating more than one response type, and a benchmark to beat.

Link: <https://www.sciencedirect.com/science/article/pii/S0140366425002099>

Theme 3 — AI-driven detection & autonomous response

Where the frontier (and most of the hype) sits: deep learning IDS, and — more relevant to us — learning-based automated response.

8. Deep reinforcement learning-based intrusion detection scheme for software-defined networking [High]

Nature Scientific Reports, 2025.

Why it matters: Representative of RL applied inside SDN. Read it for how the state/action/reward is framed — directly transferable if we make response (not detection) the RL agent's job.

Link: <https://www.nature.com/articles/s41598-025-24869-w>

9. Optimal Security Response to Network Intrusions in IT Systems [High]

K. Hammar (PhD work), *arXiv:2502.02541*, 2025.

Why it matters: Rigorous decision-theoretic framing of automated response (when to block/redirect/isolate) using control theory and game theory — the theoretical backbone our safety-constrained response engine currently lacks in the ICS literature.

Link: <https://arxiv.org/pdf/2502.02541>

10. Non-local attention enhanced deep learning for robust cyberattack detection in IIoT-based SCADA systems [High]

Nature Scientific Reports, 2026.

Why it matters: State-of-the-art SCADA detection (attention-based). Use as the detection front-end so the project can focus its novelty on response rather than re-inventing a classifier.

Link: <https://www.nature.com/articles/s41598-026-37146-1>

11. Transformers and Large Language Models for Efficient Intrusion Detection Systems: A Comprehensive Survey [Reference]

arXiv:2408.07583, 2024.

Why it matters: Breadth on transformer/LLM IDS and their cost/latency trade-offs — important because heavyweight models clash with ICS real-time budgets.

Link: <https://arxiv.org/pdf/2408.07583>

12. Design of an AI-driven secure 5G-SDN framework with federated reinforcement learning for anomaly detection, mitigation and forensics [Reference]

PMC (open access), 2025.

Why it matters: Shows the full detect-mitigate-forensics loop with federated RL over SDN — an architectural pattern to borrow, adapted to ICS constraints.

Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12929375/>

Theme 4 — In-network defense with programmable data planes (P4)

The route to line-rate enforcement that meets ICS latency budgets — a strong differentiator versus control-plane-only designs.

13. P4Control: Line-Rate Cross-Host Attack Prevention via In-Network Information Flow Control [High]

IEEE Symposium on Security and Privacy (S&P), 2024 — arXiv:2405.14970.

Why it matters: Top-tier venue proof that meaningful security policy can run at line rate on programmable switches + eBPF. Evidence that data-plane response is feasible, not just theoretical.

Link: <https://arxiv.org/pdf/2405.14970>

14. Leveraging Data Plane Programmability to enhance service orchestration at the edge: a focus on industrial security [High]

Computer Networks, Elsevier, 2024 — S1389128624002299.

Why it matters: Directly ICS/industrial-focused P4 security. Bridges programmable data planes to OT protocols and orchestration — a near-neighbour to our hybrid enforcement idea.

Link: <https://www.sciencedirect.com/science/article/pii/S1389128624002299>

15. A Two-level Intrusion Detection System for ICS Networks using P4 [High]

G. K. Ndonga, R. Sadre — ICS-CSR, 2018.

Why it matters: L1 P4 Modbus allowlist on the switch, L2 DPI feedback updates L1 — the reference design for closing the detect-to-block loop at line rate in ICS.

Link: <https://www.scienceopen.com/hosted-document?doi=10.14236/ewic/ICS2018.4>

16. A Protocol-Aware P4 Pipeline for MQTT Security and Anomaly Mitigation in Edge IoT Systems [Reference]

arXiv:2601.07536, 2026.

Why it matters: Shows protocol-aware P4 pipelines for an IIoT protocol — the pattern generalizes to Modbus/DNP3/OPC UA in our data-plane filter.

Link: <https://arxiv.org/pdf/2601.07536>

Theme 5 — Digital twins for ICS security

A fast-rising area: a safe virtual replica to test responses before applying them to the live process — a natural pairing with reactive SDN.

17. TwinSec-IDS: An Enhanced IDS in SDN-Digital-Twin-Based Industrial Cyber-Physical Systems [High]

Krishnaveni et al. — Concurrency and Computation: Practice and Experience (Wiley), 2025.

Why it matters: Explicitly fuses SDN + digital twin + hybrid deep learning for ICPS. The closest published system to a 'twin-validated reactive SDN', and therefore both inspiration and the state of the art to advance.

Link: <https://onlinelibrary.wiley.com/doi/10.1002/cpe.8334>

18. Digital Twin-Enhanced Incident Response for Cyber-Physical Systems [High]

ACM ARES, 2023.

Why it matters: Frames the twin as a sandbox for validating incident-response actions — directly supports our 'simulate the response before you commit it' safety mechanism.

Link: <https://dl.acm.org/doi/10.1145/3600160.3600195>

19. Digital Twin-Driven Intrusion Detection for Industrial SCADA: A Cyber-Physical Case Study [Reference]

MDPI Sensors, 25(16):4963, 2025.

Why it matters: Concrete SCADA case study; useful methodology and metrics for a twin-in-the-loop evaluation.

Link: <https://www.mdpi.com/1424-8220/25/16/4963>

Theme 6 — Deception & honeypots (SDN-steered)

The 'redirect for deeper inspection' branch of the project, now being supercharged with AI-generated decoy content.

20. Reactive cyber deception: stealth-based adaptive redirection to on-demand honeypots with AI-driven data generation [High]

Computer Networks / Computers & Security, Elsevier, 2026 — S138912862600215X.

Why it matters: State of the art: SDN controller performs stealthy TCP redirection to on-demand honeypots, with LLM-generated decoy data. A vivid, recent instance of exactly the redirect response our system offers — raise our game to this level.

Link: <https://www.sciencedirect.com/science/article/pii/S138912862600215X>

21. D3O-IIoT: Deep reinforcement learning-driven dynamic deception orchestration for Industrial IoT security [High]

PMC (open access), 2025.

Why it matters: RL coordinates honeypots + MTD + fake telemetry + node isolation from real-time threat signals — the multi-response orchestration idea, applied to IIoT. Strong prior art for our orchestration layer.

Link: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12816736/>

22. MimePot: a Model-based Honeypot for Industrial Control Networks [Reference]

G. Bernieri, M. Conti, F. Pascucci — IEEE SMC, 2019.

Why it matters: SDN redirects data-integrity attacks to a process-modelling honeypot — a concrete ICS honeypot design referenced by the anchor survey.

Link: <https://www.sciencedirect.com/science/article/pii/S1874548223000288>

Theme 7 — Moving Target Defense (MTD)

Proactive obfuscation that complements reactive filtering; strongest evidence base is in the smart-grid subdomain.

23. Survey of Moving Target Defense in Power Grids: Design Principles, Tradeoffs, and Future Directions [High]

arXiv:2409.18317, 2024.

Why it matters: The most complete recent MTD survey for critical infrastructure. Read for design principles and the cost/effectiveness trade-offs that any reactive-MTD trigger must respect.

Link: <https://arxiv.org/pdf/2409.18317>

24. CHAOS: An SDN-Based Moving Target Defense System [Reference]

Y. Shi et al. — *Security and Communication Networks*, 2017.

Why it matters: Widely cited SDN MTD implementation (host/port/path obfuscation) — the canonical mechanism reference.

Link: <https://www.hindawi.com/journals/scn/2017/3659167/>

25. Leveraging Network Reconfiguration to Mitigate Stealthy FDI Attacks in Smart Grid SCADA Systems

[Reference]

Springer, 2025.

Why it matters: Reactive reconfiguration that defeats a large share of stealthy false-data-injection attacks while preserving performance — a survivability response worth emulating.

Link: https://link.springer.com/chapter/10.1007/978-3-032-01904-2_19

Theme 8 — Zero Trust & micro-segmentation for OT

The dominant industry architecture direction; SDN is the natural enforcement substrate for dynamic, identity-aware segmentation.

26. The Journey to Zero Trust Microsegmentation (CISA guidance) [High]

U.S. CISA, 2025.

Why it matters: Authoritative professional/industry guidance. Grounds our device-aware policy in the zero-trust framing regulators and asset owners now expect — important for real-world relevance.

Link: https://www.cisa.gov/sites/default/files/2025-07/ZT-Microsegmentation-Guidance-Part-One_508c.pdf

27. Zero Trust Architecture: A Systematic Literature Review [Reference]

arXiv:2503.11659, 2025.

Why it matters: Academic mapping of ZTA — positions dynamic, per-flow SDN enforcement within the broader zero-trust research landscape.

Link: <https://arxiv.org/pdf/2503.11659>

Theme 9 — Deterministic security & Time-Sensitive Networking

The constraint that separates ICS from IT security: responses must not break bounded latency. TSN + SDN is the emerging way to guarantee it.

28. Software-Defined Time-Sensitive Networking for Cross-Domain Deterministic Transmission [High]

MDPI Electronics, 13(7):1246, 2024.

Why it matters: Shows how SDN and TSN combine to keep latency bounded — essential if our reactive rule changes must not violate control-loop deadlines.

Link: <https://www.mdpi.com/2079-9292/13/7/1246>

29. Time-Sensitive Networking for Industrial Automation: Current Advances and Future Directions

[Reference]

ACM Computing Surveys, 2024 — doi:10.1145/3695248.

Why it matters: Comprehensive TSN survey; use to argue and quantify the real-time envelope our security actions operate within.

Link: <https://dl.acm.org/doi/10.1145/3695248>

30. P4-PSFP: P4-Based Per-Stream Filtering and Policing for Time-Sensitive Networking [Reference]

arXiv:2311.07385, 2023.

Why it matters: Marries P4 data-plane enforcement with TSN policing — a template for security filtering that is deterministic by construction.

Link: <https://arxiv.org/pdf/2311.07385>

Theme 10 — Resilient control plane, datasets & testbeds

Making the controller trustworthy under attack, and evaluating the whole system credibly.

31. Intrusion Tolerance for Networked Systems through Two-Level Feedback Control [High]

K. Hammar, R. Stadler — *arXiv:2404.01741*, 2024.

Why it matters: Principled intrusion-tolerant control that keeps a system operating while compromised — directly relevant to keeping our reactive controller available and correct under attack.

Link: <https://arxiv.org/pdf/2404.01741>

32. ITC: Intrusion Tolerant Controller for multicontroller SDN architecture [Reference]

Computers & Security, Elsevier, 2023 — S0167404823002614.

Why it matters: Concrete intrusion-tolerant SDN controller design; the resilience building block for a reactive control plane.

Link: <https://www.sciencedirect.com/science/article/abs/pii/S0167404823002614>

33. SWaT, WADI and HAI ICS security datasets / testbeds (iTrust, Singapore) [Essential]

Secure Water Treatment (SWaT), Water Distribution (WADI), HIL-Augmented ICS (HAI).

Why it matters: The de-facto benchmarks for ICS security. Use for repeatable, comparable evaluation; note their limitation (offline, detection-oriented) which motivates our closed-loop testbed.

Link: https://itrust.sutd.edu.sg/itrust-labs_datasets/

34. MiniCPS: A Toolkit for Security Research on CPS Networks [Reference]

D. Antonoli, N. O. Tippenhauer — *ACM CPS-SPC*, 2015.

Why it matters: The emulation backbone (used by Piedrahita et al.) for building a Ryu + Mininet + Modbus reactive testbed with realistic physical-process behaviour.

Link: <https://dl.acm.org/doi/10.1145/2808705.2808715>

Literature & market gap analysis

Across all ten themes a consistent asymmetry appears. An enormous, still-growing body of work — much of it reporting near-perfect accuracy on SWaT/WADI/HAI — is devoted to *detecting* attacks. Far less work addresses the harder, higher-value question our project targets: given a detection, what network action should the controller take, and how do we guarantee that action does not itself harm the physical process? The professional/industry signal points the same way — OT ransomware rose ~87% in 2024 and internet-exposed ICS devices keep climbing, so asset owners increasingly need automated *response*, not more alerts. The table below summarizes the white space.

Well covered in the literature	Under-served gap	Opportunity for us
Detecting attacks (ML/DL/GNN IDS on SWaT/WADI); very high reported accuracy.	Turning a detection into a SAFE, automated network action that respects the physical process.	Safety-constrained response engine.
SDN blocking / redirection / honeypot primitives demonstrated individually.	A unified controller that ingests events from ANY external source and selects device-aware responses.	Source-agnostic event API + policy engine.
Honeypots & MTD as standalone tactics; LLM-enriched deception	Orchestrating multiple responses (block, mirror, redirect, re-route, deceive) under one	Response orchestration layer.

Well covered in the literature	Under-served gap	Opportunity for us
emerging.	risk-aware policy.	
Control-plane logic and cloud/NFV analytics.	Line-rate enforcement on the switch (P4) coupled to slower controller reasoning.	Hybrid data-plane / control-plane loop.
Controller single-point-of-failure noted; BFT/multi-controller in generic SDN.	Reactive ICS response that stays available and correct while the controller itself is attacked.	Resilient reactive control plane.
High accuracy on offline datasets.	Real-time closed-loop evaluation with detection-to-mitigation latency and safety metrics.	Reproducible closed-loop testbed.

Five gaps stand out as genuinely under-served and defensible:

- **The response gap.** Detection is commoditized; safe, automated, reversible response in ICS is not. Only Piedrahita et al. and a handful of others treat process safety as a first-class constraint on the action.
- **Source-agnostic orchestration.** Systems bind to one detector and one response. A controller that ingests standardized events from any external service and orchestrates block / mirror / redirect / re-route / deceive under one risk-aware policy does not yet exist for ICS.
- **Hybrid data-plane / control-plane loops.** P4 line-rate enforcement and controller-level reasoning are studied separately; coupling fast in-network mitigation to slower, smarter controller decisions is largely open.
- **Twin-validated, safety-constrained automation.** Digital twins are used for detection, rarely to pre-validate a response before it touches the live process — a concrete way to make automation trustworthy.
- **Closed-loop, real-time evaluation.** The field evaluates on offline datasets; detection-to-mitigation latency, process-safety preservation, and controller resilience under attack are seldom measured together.

Where we can innovate — differentiated directions

Synthesizing the gaps into buildable, publishable contributions for this project:

- **A safety-constrained reactive response engine.** Formalize response selection (à la Hammar's optimal-response work) with an explicit ICS safety envelope, so an automated action is only applied if it provably keeps the process within safe bounds — otherwise it degrades gracefully (mirror/redirect instead of block).
- **A source-agnostic event-to-action API + policy engine on Ryu.** Standardize the interface so any IDS/asset-discovery/anomaly service can drive the controller, and make responses device-aware (block an unseen device; redirect a critical service) — the exact behaviour in the project's problem statement, generalized.
- **A hybrid enforcement loop.** Push allowlist/rate-limit filtering to a P4 data plane for line-rate, deterministic mitigation, while the Ryu controller performs slower risk-aware reasoning and updates the fast path — meeting TSN-style latency budgets.

- **Twin-in-the-loop response validation.** Before committing a disruptive action, simulate it on a lightweight digital twin/MiniCPS model and apply only if the predicted process state stays safe — turning digital twins from a detection aid into a safety gate.
- **Orchestrated, escalating deception.** Extend redirect-to-honeypot with on-demand, LLM-enriched decoys, escalated by policy — combining Themes 3 and 6 into a graduated response ladder rather than a binary allow/block.
- **A resilient reactive control plane + a closed-loop benchmark.** Pair the reactive logic with intrusion-tolerant control so it survives attacks on the controller itself, and publish a reproducible testbed that measures detection-to-mitigation latency, safety preservation, and resilience together — a contribution the community currently lacks.

Highest-leverage single thesis: a safety-constrained, source-agnostic reactive SDN controller on Ryu that validates each response against a digital twin and enforces at the P4 data plane. It sits precisely in the intersection of the five gaps above — novel, feasible on the existing Ryu/Mininet/MiniCPS toolchain, and directly aligned with the project's stated objective.

Suggested reading order

If time is limited, read in this sequence: (1) the Etxezarreta survey for the map; (2) Piedrahita et al. for the closest prior system; (3) the 2025 ML/DL-for-SDN systematic review for the AI frontier; (4) Hammar's optimal-response work for the theory of safe response; (5) P4Control and the industrial-P4 paper for line-rate enforcement; (6) TwinSec-IDS and the ARES digital-twin incident-response paper for twin-in-the-loop; (7) the reactive-deception paper for the redirect branch; then skim the remaining themes as needed. Everything above is captured in the project README status so you can track what you have read.

Sourcing note: entries were gathered via a structured web sweep across ten sub-domains (July 2026). Venue/year are given as found; verify exact bibliographic details against the publisher of record before formal citation. A few items are attributed via the Etxezarreta et al. (2023) survey where the original is paywalled.